Gokul M

AI/ML Engineer | Machine Learning Engineer | Generative AI Engineer
Phone: +91 8098403166 | Email: gokulmani200429@gmail.com | LinkedIn: linkedin.com/in/gokul-m-0505a4279
GitHub: github.com/Gokulm29 | Location: Chennai, Tamil Nadu

About Me

AI/ML Engineer with hands-on experience in building and deploying end-to-end AI solutions to solve real-world problems. Developed four AI systems with live demonstrations and public GitHub repositories, focusing on practical applications, scalability, and reliability. Skilled in Python, Machine Learning, Deep Learning, and Generative AI, with a strong interest in developing production-ready AI applications and contributing to engineering teams.

Education

Karpagam College of Engineering — Coimbatore, India
B.Tech Artificial Intelligence and Data Science (CGPA: 7.60) — *2022 – 2026*

S.B.K Hr.Sec School — State Board
Higher Secondary Certificate (56.67%) — *2022*

S.B.K Hr.Sec School — State Board
Secondary School Leaving Certificate (63.00%) — *2020*

Projects

RAG-Based AI/ML Interview Assistant — Oct 2026

- Developed a Retrieval-Augmented Generation (RAG) chatbot using Python that retrieves relevant content from interview preparation PDFs and generates context-aware answers to technical AI/ML questions
- Built the data pipeline for PDF text extraction, text chunking, and sentence embeddings with Sentence Transformers, and implemented semantic search over a FAISS vector database
- Integrated the Groq LLM API (openai/gpt-oss-120b) for Generative AI responses grounded in retrieved context, with source references and interactive Chat and Mock Interview modes
- Deployed the web application on Streamlit Community Cloud for real-time AI/ML interview preparation
- Technologies: Python, RAG, FAISS, Sentence Transformers, Groq API, LLM, PyPDF, Git, GitHub, Streamlit Community Cloud

AI-Powered Multi-Agent Procurement System for Supply Chain Automation — Aug 2025 - Mar 2026

- Developed an end-to-end multi-agent system (12 agents) to automate the procurement lifecycle
- Implemented ML-based demand forecasting and intelligent inventory-based replenishment
- Built automated supplier discovery, RFQ generation, and quote analysis system
- Integrated LLM-based document verification and decision-making with PDF report generation
- Technologies: Python, LangChain, CrewAI, Groq API, Streamlit, Pandas, NumPy, FAISS, ChromaDB, ReportLab, Github

FIFA Analytics Dashboard — Jul 2024

- Designed and developed an interactive football analytics dashboard using Tableau.
- Enabled visual insights for player and team performance analysis
- Technologies: Tableau, Data Visualization, Dashboard Design, Github

MLOps for Job Placement Prediction — Feb 2024 – Mar 2024

- Developed an end-to-end machine learning pipeline for predicting student job placement using a real-world Kaggle dataset.
- Performed data preprocessing, feature engineering, model training, and evaluation, selecting the Random Forest model with 86.05% Accuracy, 90.32% F1-Score, and 0.93 ROC-AUC.
- Built an interactive Streamlit web application for real-time placement prediction, serialized the trained model using Joblib, and containerized the application using Docker for consistent deployment across different environments.
- Technologies: Python, Scikit-learn, Pandas, NumPy, Streamlit, Docker, Joblib, Git, GitHub.

Internship Experience

Octanet Services — Sep 2023 – Oct 2023
Python Developer Intern

- Designed and implemented an ATM Management System using Python and Object-Oriented Programming principles.
- Improved transaction flow and system performance

Skills

Languages: Python, SQL
Libraries & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras
Machine Learning & MLOps: Regression, Classification, Clustering, Exploratory Data Analysis (EDA), Feature Engineering, Text Preprocessing, Model Evaluation, Hyperparameter Tuning, Model Deployment, Model Serialization
Deep Learning: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Artificial Neural Networks (ANN)
Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, CrewAI, Groq API, Embeddings, FAISS, ChromaDB
Deployment: Streamlit, Docker, Joblib
Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Tableau

Certifications

Python Development Internship - Octanet services — Oct 2023
Data Analytics With Python - NPTEL — Apr 2024
Data Visualization with Power BI - Great Learning — Aug 2024
Introduction to MongoDB - MongoDB — Sep 2024
Business Analysts Qualification - Qlik Sense — Oct 2024